

Quantum Systems

(Lecture 3: The principles of quantum computation I)

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The principles

Quantum computation explores the laws of quantum theory as computational resources.

Thus, the principles of the former are directly derived from the postulates of the latter.

- The state representation postulate
- The state evolution postulate
- The state composition postulate
- The state measurement postulate

The underlying maths is that of Hilbert spaces.

The underlying maths: Hilbert spaces

Complex, inner-product vector space

A complex vector space with **inner product**

$$\langle - | - \rangle : V \times V \longrightarrow \mathbb{C}$$

such that

$$(1) \quad \langle v | \sum_i \lambda_i \cdot |w_i\rangle \rangle = \sum_i \lambda_i \langle v | w_i \rangle$$

$$(2) \quad \langle v | w \rangle = \overline{\langle w | v \rangle}$$

$$(3) \quad \langle v | v \rangle \geq 0 \text{ (with equality iff } |v\rangle = 0)$$

Note: $\langle - | - \rangle$ is **conjugate linear** in the first argument:

$$\langle \sum_i \lambda_i \cdot |w_i\rangle | v \rangle = \sum_i \overline{\lambda_i} \langle w_i | v \rangle$$

Notation: $\langle v | w \rangle \equiv \langle v, w \rangle \equiv (|v\rangle, |w\rangle)$

Dirac's notation

Dirac's bra/ket notation is a handy way to represent elements and constructions on an Hilbert space, amenable to calculations and with direct correspondence to diagrammatic representations of process theories

$|u\rangle$ A **ket** stands for a vector in an Hilbert space V . In $\mathbb{C}^n$, a column vector of complex entries. The identity for $+$ (the **zero** vector) is just written 0 .

$\langle u|$ A **bra** is a vector in the **dual** space $V^\dagger$, i.e. scalar-valued linear maps in V — a row vector in $\mathbb{C}^n$.

There is a bijective correspondence between $|u\rangle$ and $\langle u|$

$$|u\rangle = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} \Leftrightarrow [\bar{u}_1 \cdots \bar{u}_n] = \langle u|$$

Inner product: examples

In $\mathbb{C}$

$$\langle a + bi | c + di \rangle = (a - bi)(c + di) = ac + adi - bci + bd$$

In $\mathbb{C}^n$: The dot product

A useful example of a **inner product** is the **dot product**

$$\langle u | v \rangle = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \cdot \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \underbrace{[\overline{u_1} \quad \overline{u_2} \quad \cdots \quad \overline{u_n}]}_{\langle u |} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \sum_{i=1}^n \overline{u_i} v_i$$

where $\bar{c} = a - ib$ is the complex conjugate of $c = a + ib$

$\langle u |$ is the **adjoint** of vector $|u\rangle$, i.e a vector in the **dual** vector space $V^\dagger$.

The dual space

 $V^\dagger$

If V is a Hilbert space, $V^\dagger$ is the space of **linear maps** from V to $\mathbb{C}$.

Elements of $V^\dagger$ are denoted by

$$\langle u| : V \longrightarrow \mathbb{C} \text{ defined by } \langle u|(|v\rangle) = \langle u|v\rangle$$

In a matricial representation $\langle u|$ is obtained as the **Hermitian conjugate** (i.e. the **transpose** of the vector composed by the **complex conjugate** of each element) of $|u\rangle$, therefore the dot product of $|u\rangle$ and $|v\rangle$.

The adjoint operator

Given an operator $U : H \longrightarrow H$, its **adjoint** $U^\dagger : H^\dagger \longrightarrow H^\dagger$ is the unique operator satisfying

$$\langle U^\dagger w | v \rangle = \langle w | Uv \rangle \quad (1)$$

an equality that is often denoted by expression

$$\langle u | U | v \rangle$$

Note that $(UV)^\dagger = V^\dagger U^\dagger$ because

$$\begin{aligned} \langle (UV)^\dagger w | v \rangle &= \langle w | UVv \rangle \\ &= \langle U^\dagger w | Vv \rangle \\ &= \langle V^\dagger U^\dagger w | v \rangle \end{aligned}$$

The matrix representation of $U^\dagger$ is the conjugate transpose of that of U

Exercise: Prove that $\overline{\langle w | U | v \rangle} = \langle v | U^\dagger | w \rangle$

Old friends: Norms and orthogonality

Old friends

- $|v\rangle$ and $|w\rangle$ are **orthogonal** if $\langle v|w\rangle = 0$
- **norm**: $\| |v\rangle \| = \sqrt{\langle v|v\rangle}$, a nonnegative real number

This norm satisfies $\| |v\rangle + |w\rangle \| \leq \| |v\rangle \| + \| |w\rangle \|$ due to the Cauchy-Schwarz inequality:

$$\langle x|y\rangle^2 \leq \langle x|x\rangle \langle y|y\rangle$$

Old friends: Norms and orthogonality

Old friends

- **normalization:** $\frac{|v\rangle}{\|v\|}$
- $|v\rangle$ is a **unit vector** if $\| |v\rangle \| = 1$
- A set of vectors $\{|i\rangle, |j\rangle, \dots, \}$ is **orthonormal** if each $|i\rangle$ is a unit vector and

$$\langle i | j \rangle = \delta_{i,j} = \begin{cases} i = j & \Rightarrow 1 \\ \text{otherwise} & \Rightarrow 0 \end{cases}$$

Old friends: Bases

Orthonormal basis

A orthonormal basis for a Hilbert space V of dimension n is a set $B = \{|i\rangle\}$ of n linearly independent elements of V st

- $\langle i|j\rangle = \delta_{i,j}$ for all $|i\rangle, |j\rangle \in B$
- and B **spans** V , i.e. every $|v\rangle$ in V can be written as

$$|v\rangle = \sum_i \alpha_i |i\rangle \quad \text{for some } \alpha_i \in \mathbb{C}$$

Note that the **amplitude** or **coefficient** of $|v\rangle$ wrt $|i\rangle$ satisfies

$$\alpha_i = \langle i|v\rangle$$

Why?

Old friends: Bases

$\alpha_i = \langle i | v \rangle$ because

$$\begin{aligned}\langle i | v \rangle &= \langle i | \sum_j \alpha_j | j \rangle \\ &= \sum_j \alpha_j \langle i | j \rangle \\ &= \sum_j \alpha_j \delta_{i,j} \\ &= \alpha_i\end{aligned}$$

Note

If $|v\rangle$ is expressed wrt any orthonormal basis $\{|i\rangle\}$, i.e. $|v\rangle = \sum_i \alpha_i |i\rangle$, then

$$\| |v\rangle \| = \sum_i \| \alpha_i \|^2$$

Example: The Hadamard basis

One of the infinitely many orthonormal bases for a space of dimension 2:

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

Check e. g.

$$\langle + | - \rangle = \frac{1}{2}(\langle 0 | + \langle 1 |, |0\rangle - |1\rangle) = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 0$$

$$\| |+\rangle \| = \sqrt{\langle + | + \rangle} = \sqrt{\frac{1}{2}(\langle 0 | + \langle 1 |, |0\rangle + |1\rangle)} = \sqrt{\frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}} = 1$$

Bases

A basis for $V^\dagger$

If $\{|i\rangle\}$ is an orthonormal basis for V , then

$$\{\langle i|\}$$

is an orthonormal basis for $V^\dagger$

Hilbert spaces

The complete picture

An **Hilbert space** is an inner-product space V st the metric defined by its norm turns V into a **complete metric space**, i.e. any Cauchy sequence

$$|v_1\rangle, |v_2\rangle, \dots$$

$$\forall \epsilon > 0 \exists N \forall_{m,n > N} \quad ||v_m - v_n|| \leq \epsilon$$

converges

(i.e. there exists an element $|s\rangle$ in V st $\forall \epsilon > 0 \exists N \forall_{n > N} \quad ||s - v_n|| \leq \epsilon$)

The completeness condition is trivial in **finite dimensional** vector spaces

The representation postulate

Postulate 1

The state space of a quantum system is described by a unit vector in a Hilbert space

- In practice, with finite resources, one cannot distinguish between a **continuous** state space from a **discrete** one with arbitrarily small minimum spacing between adjacent locations.
- One may, then, restrict to **finite-dimensional** (complex) Hilbert spaces.

The representation postulate

A quantum (binary) state is represented as a **superposition**, i.e. a linear combination of vectors $|0\rangle$ and $|1\rangle$ with **complex** coefficients:

$$|\phi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

When state $|\phi\rangle$ is **measured** (i.e. **observed**) one of the two basic states $|0\rangle, |1\rangle$ is returned with probability

$$\|\alpha\|^2 \quad \text{and} \quad \|\beta\|^2$$

respectively.

Being probabilities, the norm squared of coefficients must satisfy

$$\|\alpha\|^2 + \|\beta\|^2 = 1$$

which enforces quantum states to be represented by **unit** vectors.

The state space of a qubit

Global phase

Unit vectors equivalent up to multiplication by a complex number of modulus one, i.e. a **phase factor** $e^{i\theta}$, represent the **same** state.

Let

$$|v\rangle = \alpha|u\rangle + \beta|u'\rangle$$

$$\|e^{i\theta}\alpha\|^2 = (\overline{e^{i\theta}\alpha})(e^{i\theta}\alpha) = (e^{-i\theta}\overline{\alpha})(e^{i\theta}\alpha) = \overline{\alpha}\alpha = \|\alpha\|^2$$

and similarly for β .

As the probabilities $\|\alpha\|^2$ and $\|\beta\|^2$ are the **only** measurable quantities, **global phase has no physical meaning**.

Representation redundancy

qubit state space $\neq$ complex vector space used for representation

The state space of a qubit

Relative phase

It is a measure of the angle between the two complex numbers.
Thus, it cannot be discarded!

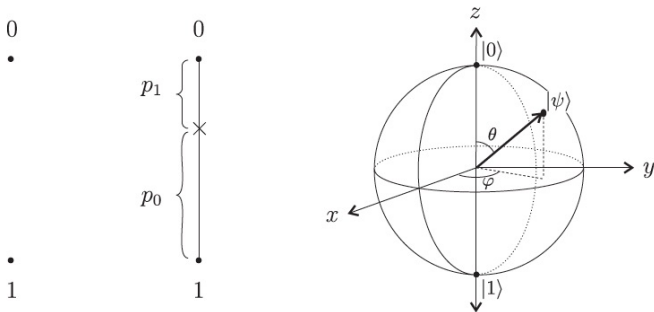
Those are different states

$$\frac{1}{\sqrt{2}}(|u\rangle + |u'\rangle) \quad \frac{1}{\sqrt{2}}(|u\rangle - |u'\rangle) \quad \frac{1}{\sqrt{2}}(e^{i\theta}|u\rangle + |u'\rangle)$$

...

The Bloch sphere

Deterministic, probabilistic and quantum bits



(from [Kaeys et al, 2007])

The Bloch sphere: Representing $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

- Express $|\psi\rangle$ in **polar** form

$$|\psi\rangle = \rho_1 e^{i\varphi_1} |0\rangle + \rho_2 e^{i\varphi_2} |1\rangle$$

- Eliminate one of the four real parameters multiplying by $e^{-i\varphi_1}$

$$|\psi\rangle = \rho_1 |0\rangle + \rho_2 e^{i(\varphi_2 - \varphi_1)} |1\rangle = \rho_1 |0\rangle + \rho_2 e^{i\varphi} |1\rangle$$

making $\varphi = \varphi_2 - \varphi_1$,

which is possible because **global phase factors** are **physically meaningless**.

The Bloch sphere: Representing $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

- Switching back the coefficient of $|1\rangle$ to Cartesian coordinates

$$|\psi\rangle = \rho_1|0\rangle + (a + bi)|1\rangle$$

the normalization constraint

$$\|\rho_1\|^2 + \|a+ib\|^2 = \|\rho_1\|^2 + (a-ib)(a+ib) = \boxed{\|\rho_1\|^2 + a^2 + b^2 = 1}$$

yields the [equation of a unit sphere](#) in the real tridimensional space with Cartesian coordinates: (a, b, ρ_1) .

The Bloch sphere: Representing $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

- The **polar** coordinates (ρ, θ, φ) of a point in the surface of a sphere relate to Cartesian ones (x, y, z) through the correspondence

$$x = \rho \sin \theta \cos \varphi$$

$$y = \rho \sin \theta \sin \varphi$$

$$z = \rho \cos \theta$$

- Recalling $\rho = 1$ (cf unit sphere),

$$\begin{aligned} |\psi\rangle &= \rho_1|0\rangle + (a + ib)|1\rangle \\ &= \cos \theta |0\rangle + \sin \theta (\cos \varphi + i \sin \varphi) |1\rangle \\ &= \cos \theta |0\rangle + e^{i\varphi} \sin \theta |1\rangle \end{aligned}$$

which, with **two parameters**, defines a **point** in the sphere's surface.

The Bloch sphere

Actually, one may just focus on the **upper hemisphere** ($0 \leq \theta' \leq \frac{\pi}{2}$) as opposite points in the lower one differ only by a phase factor of -1 , as suggested by

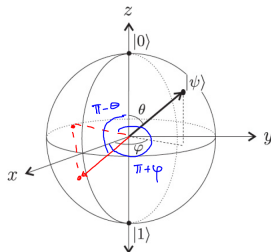
$$\theta' = 0 \Rightarrow |\psi\rangle = \cos 0|0\rangle + e^{i\varphi} \sin 0|1\rangle = |0\rangle$$

$$\theta' = \frac{\pi}{2} \Rightarrow |\psi\rangle = \cos \frac{\pi}{2}|0\rangle + e^{i\varphi} \sin \frac{\pi}{2}|1\rangle = e^{i\varphi}|1\rangle = |1\rangle$$

Note that **longitude** (φ) is irrelevant in a pole!

The Bloch sphere

Indeed, let $|\psi'\rangle$ be the opposite point on the sphere with polar coordinates $(1, \pi - \theta, \varphi + \pi)$:



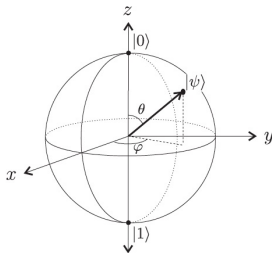
$$\begin{aligned}
 |\psi'\rangle &= \cos(\pi - \theta)|0\rangle + e^{i(\varphi + \pi)} \sin(\pi - \theta)|1\rangle \\
 &= -\cos\theta|0\rangle + e^{i\varphi} e^{i\pi} \sin\theta|1\rangle \\
 &= -\cos\theta|0\rangle + e^{i\varphi} \sin\theta|1\rangle \\
 &= -|\psi\rangle
 \end{aligned}$$

The Bloch sphere

which leads to

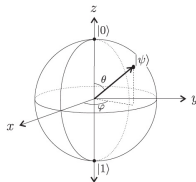
$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle$$

where $0 \leq \theta \leq \pi$, $0 \leq \varphi \leq 2\pi$



The map $\frac{\theta}{2} \mapsto \theta$ is **one-to-one** at any point but at $\frac{\theta}{2}$:
all points on the equator are mapped into a single point: the south pole.

The Bloch sphere



- The poles represent the classical bits. In general, **orthogonal states correspond to antipodal points** and every **diameter** to a **basis** for the single-qubit state space.
- Once measured a qubit collapses to one of the two poles. Which pole depends exactly on the arrow direction: The angle θ measures that **probability**: If the arrow points at the equator, there is 50-50 chance to collapse to any of the two poles.
- Rotating a vector wrt the z -axis results into a **phase change** (φ), and does not affect which state the arrow will collapse to, when measured.

End of parenthesis

...)

The state evolution postulate

If a quantum state is a **ray** (i.e. a unit vector in a Hilbert space H up to a global phase), its evolution is specified a certain kind of **linear** operators $U: H \longrightarrow H$.

Linearity

$$U \left(\sum_j \alpha_j |v_j\rangle \right) = \sum_j \alpha_j U(|v_j\rangle)$$

just by itself has an important consequence: **quantum states cannot be cloned**

The no-cloning theorem

Linearity implies that quantum states cannot be cloned

Let $U(|a\rangle|0\rangle) = |a\rangle|a\rangle$ be a 2-qubit operator and $|c\rangle = \frac{1}{\sqrt{2}}(|a\rangle + |b\rangle)$ for $|a\rangle, |b\rangle$ orthogonal. Then,

$$\begin{aligned}U(|c\rangle|0\rangle) &= \frac{1}{\sqrt{2}}(U(|a\rangle|0\rangle) + U(|b\rangle|0\rangle)) \\&= \frac{1}{\sqrt{2}}(|a\rangle|a\rangle + |b\rangle|b\rangle) \\&\neq \frac{1}{\sqrt{2}}(|a\rangle|a\rangle + |a\rangle|b\rangle + |b\rangle|a\rangle + |b\rangle|b\rangle) \\&= |c\rangle|c\rangle \\&= U(|c\rangle|0\rangle)\end{aligned}$$

As already seen, $|x\rangle|y\rangle = |xy\rangle = |x\rangle \otimes |y\rangle$

The state evolution postulate

Postulate 2

The evolution over time of the state of a closed quantum system is described by a unitary operator.

The evolution is **linear**

$$U \left(\sum_j \alpha_j |v_j\rangle \right) = \sum_j \alpha_j U(|v_j\rangle)$$

and preserves the **normalization constraint**

$$\text{If } \sum_j \alpha_j U(|v_j\rangle) = \sum_j \alpha'_j |v_j\rangle \text{ then } \sum_j \|\alpha'_j\|^2 = 1$$

The state evolution postulate

Preservation of the **normalization constraint** means that unit length vectors (and thus orthogonal subspaces) are mapped by U to unit length vectors (and thus to orthogonal subspaces).

This entails a condition on valid quantum operators: they must **preserve** the inner product, i.e.

$$\langle U|v\rangle, U|w\rangle \rangle = \langle v|U^\dagger U|w\rangle = \langle v|w\rangle$$

which is the case iff U is **unitary**, i.e. $U^\dagger = U^{-1}$:

$$U^\dagger U = UU^\dagger = I$$

Unitarity

- Preserving the inner product means that a unitary operator maps **orthonormal bases** to **orthonormal bases**.
- Conversely, any operator with this property is unitary.
- If given in matrix form, being unitary means that the set of columns of its matrix representation are orthonormal (because the j th column is the image of $U|j\rangle$). Equivalently, rows are orthonormal (why?)

Unitarity

Unitarity is the **only** constraint on quantum operators: Any unitary matrix specifies a valid quantum operator.

This means that there are many non-trivial operators on a single qubit (in contrast with the **classical** case where the only non-trivial operation on a bit is **complement**).

Finally, because the **inverse** of a unitary matrix is also a unitary matrix, a quantum operator can always be inverted by another quantum operator

Unitary transformations are reversible

Building larger states from smaller

Operator U in the no-cloning theorem acts on a 2-dimensional state, i.e. over the composition of two qubits.

What does composition mean?

Postulate 3

The state space of a combined quantum system is the tensor product $V \otimes W$ of the state spaces V and W of its components.

Composing quantum states

State spaces in a **quantum** system combine through **tensor**: $\otimes$

n m -dimensional vectors $\rightsquigarrow$ a vector in m^n -dimensional space

i.e. the state space of a quantum system grows exponentially with the number of particles: cf, Feynman's original motivation

Example

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} \otimes \begin{bmatrix} d \\ e \\ f \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \begin{bmatrix} d \\ e \\ f \end{bmatrix} = \begin{bmatrix} ad \\ ae \\ af \\ bd \\ be \\ bf \\ cd \\ ce \\ cf \end{bmatrix}$$

Composing quantum states

Tensor $V \otimes W$

- $B_{V \otimes W}$ is a set of elements of the form $|v_i\rangle \otimes |w_j\rangle$, for each $|v_i\rangle \in B_V$, $|w_j\rangle \in B_W$ and $\dim(V \otimes W) = \dim(V) \times \dim(W)$
- $(|u_1\rangle + |u_2\rangle) \otimes |z\rangle = |u_1\rangle \otimes |z\rangle + |u_2\rangle \otimes |z\rangle$
- $|z\rangle \otimes (|u_1\rangle + |u_2\rangle) = |z\rangle \otimes |u_1\rangle + |z\rangle \otimes |u_2\rangle$
- $(\alpha|u\rangle) \otimes |z\rangle = |u\rangle \otimes (\alpha|z\rangle) = \alpha(|u\rangle \otimes |z\rangle)$
- $\langle (|u_2\rangle \otimes |z_2\rangle) | (|u_1\rangle \otimes |z_1\rangle) \rangle = \langle u_2 | u_1 \rangle \langle z_2 | z_1 \rangle$

Composing quantum states

Clearly, every element of $V \otimes W$ can be written as

$$\alpha_1(|v_1\rangle \otimes |w_1\rangle) + \alpha_2(|v_2\rangle \otimes |w_1\rangle) + \cdots + \alpha_{nm}(|v_n\rangle \otimes |w_m\rangle)$$

Example

The basis of $V \otimes W$, for V, W qubits with the computational basis is

$$\{|0\rangle \otimes |0\rangle, |0\rangle \otimes |1\rangle, |1\rangle \otimes |0\rangle, |1\rangle \otimes |1\rangle\}$$

Thus, the tensor of $\alpha_1|0\rangle + \alpha_2|1\rangle$ and $\beta_1|0\rangle + \beta_2|1\rangle$ is

$$\alpha_1\beta_1|0\rangle \otimes |0\rangle + \alpha_1\beta_2|0\rangle \otimes |1\rangle + \alpha_2\beta_1|1\rangle \otimes |0\rangle + \alpha_2\beta_2|1\rangle \otimes |1\rangle$$

i.e., in a simplified notation,

$$\alpha_1\beta_1|00\rangle + \alpha_1\beta_2|01\rangle + \alpha_2\beta_1|10\rangle + \alpha_2\beta_2|11\rangle$$

Bases

The computational basis for a vector space

$$\underbrace{V \otimes V \otimes \dots \otimes V}_n$$

corresponding to the composition of n qubits (each living in V) is the set

$$\begin{aligned} & \{ \underbrace{|0\rangle \dots |0\rangle |0\rangle}_n, \underbrace{|0\rangle \dots |0\rangle |1\rangle}_n, \underbrace{|0\rangle \dots |1\rangle |0\rangle}_n, \dots, \underbrace{|1\rangle \dots |1\rangle |1\rangle}_n \} \\ \stackrel{\text{abv}}{=} & \{ \underbrace{|0 \dots 00\rangle}_n, \underbrace{|0 \dots 01\rangle}_n, \underbrace{|0 \dots 10\rangle}_n, \dots, \underbrace{|1 \dots 11\rangle}_n \} \end{aligned}$$

which may be written in a compressed (decimal) way as

$$\{|0\rangle, |1\rangle, |2\rangle, |3\rangle, \dots, |2^n - 1\rangle\}$$

Bases

The **computational basis** for a two qubit system would be

$$\{|0\rangle, |1\rangle, |2\rangle, |3\rangle\}$$

with

$$|0\rangle = |00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad |1\rangle = |01\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad |2\rangle = |10\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad |3\rangle = |11\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Bases

There are of course other bases ... besides the **standard** one, e.g.

The Bell basis

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

Compare with the Hadamard basis for the single qubit systems

Representing multi-qubit states

Any unit vector in a 2^n Hilbert space represents a possible n -qubit state, but for

... a certain level of redundancy

- As before, vectors that differ only in a **global phase** represent the **same** quantum state
- but also the **same phase factor in different qubits** of a tensor product represent the **same** state:

$$|u\rangle \otimes (e^{i\phi}|z\rangle) = e^{i\phi}(|u\rangle \otimes |z\rangle) = (e^{i\phi}|u\rangle) \otimes |z\rangle$$

Actually, phase factors in qubits of a single term of a superposition can always be factored out into a coefficient for that term, i.e.
phase factors distribute over tensors